NEWTON LAWS OF MOTION

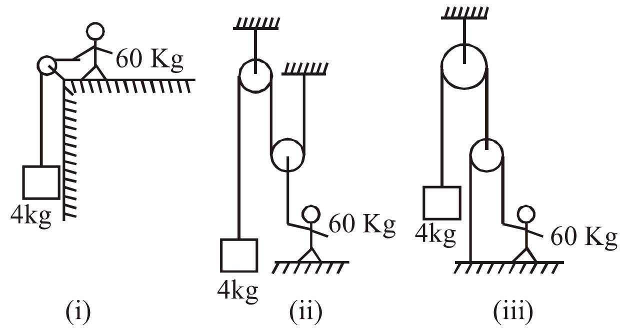
1. Consider the following statements :-
(A) Rolling friction can be smaller than static or sliding friction even by 2 or 3 times of magnitude.
(B) Lubricants are way of reducing static friction in a machine.
(C) Friction involved in ball bearings and surface in contact is rolling friction.
(D) A thin cushion of air maintained between solid surfaces in contact reduces friction.
The correct statements from above are :
(1) A, B, C (2) A, B, C, D
(3) A, C, D (4) B, C, D

2. The difference between fundamental laws and empirical relations is that empirical relations are approximately true.
Which of the given are empirical relations :-
(1) Law of gravitation (2) Coulomb's Law
(3) Laws of friction (4) None of these

3. Which of the following statements are true ?
(A) For applying the second law of motion, there is no conceptual distinction between inanimate and animate objects
(B) In merry-go-round, all parts of body are subject to an inward force and the direction of impending motion is inwards too.
(C) During the action of impulsive force there is no appreciable change in momentum of body.
(D) During the action of impulsive force there is no appreciable change in the position of body.
(1) A, B, C, D (2) B, C, D
(3) A, D (4) A, B, D

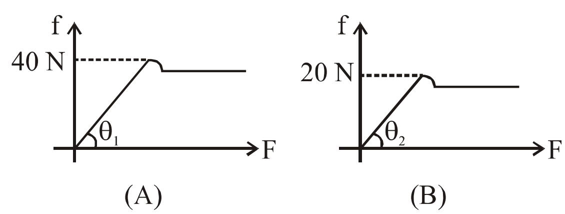
4. Consider the following statements and select incorrect statements :-
(A) The second law of newton is a vector law.
(B) The second law is a local law i.e, acceleration does not depend on the history of motion.
(C) The second law is applicable only for point objects.
(D) Only conservative forces are written in newton's second law.
(1) Only B (2) Only C
(3) B, D (4) C, D

5. Consider three cases in equilibrium :-



A man of mass 60 kg is holding the string connected to block of mass 4kg as shown. Then the incorrect statement is :-
(1) Normal reaction will be maximum on man in first case.
(2) Normal reaction will be least in third case.
(3) Tension in the string held by man is least in third case
(4) Tension in string held by man is maximum in second case.

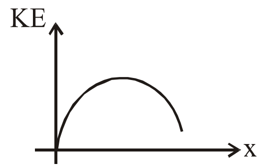
6. If the two curves drawn between frictional force f and applied force F for 2 bodies on ground. Then correct statement is :-



(1) $\theta_1 > \theta_2$
(2) Coefficient of static friction may be more for (B)
(3) Coefficient of kinetic friction is same as coefficient of static friction for both (A) & (B)
(4) Coefficient of static friction will be more for (A)

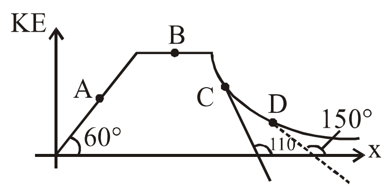
7. Conservation of momentum in a collision can be understood from :-
(1) Newton's first law only
(2) Conservation of energy principle
(3) Newton's second law only
(4) Newton's third law

8. In the given KE vs position (x) graph.



(1) Acceleration is constant
(2) Net force is increasing
(3) Magnitude of net force first increases then decreases.
(4) Magnitude of net force first decreases then increases.

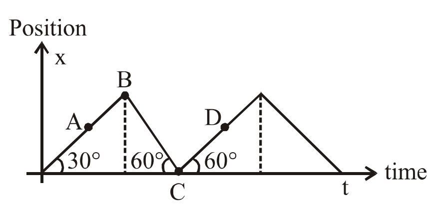
9. At which point in the given graph, magnitude of rate of change of momentum is maximum.



(1) A (2) B (3) C (4) D

10. Total number of correct statements are :-
(A) Tension may be impulsive force
(B) Gravitational force may be impulsive force
(C) Spring force can never be impulsive force
(D) Frictional force can never be impulsive
(1) 1 (2) 2 (3) 3 (4) Zero

11.



The correct ascending order of magnitude of impulse at points A, B, C, D is :-
(1) A < B < C < D (2) A = B < C < D
(3) A = D < B < C (4) A = D < C < B

12. If $v = P + qt^2 + rt^3$. Then net force acting on this particle will be zero at t = 2 if :

(1) $\left|\frac{p}{q}\right| = 0$ (2) $\frac{q}{r} = 3$

(3) $\left|\frac{r}{q}\right| = \frac{1}{3}$ (4) $\left|\frac{r}{q}\right| = \frac{-1}{3}$

13. If same force acts on different bodies (Both bodies are moving with different velocities) for same time, then :
(1) Change in speed is same for each.
(2) Change in momentum is same for each.
(3) Change in position will be same for each.
(4) Acceleration will be same for each.

14. According to Galileo's experiment for a double incline plane, if slope of second plane is decreased and the planes are smooth, then a ball is released from rest on one of the planes rolls down and climb up the other the final height of the ball is :-
(1) Less than initial height
(2) More than initial height
(3) Equal to initial height
(4) More or less than initial height

15. A ball is travelling with uniform translatory motion. This means that :-
(1) It is at rest
(2) The path can be straight line or circular and the ball travels with uniform speed
(3) All parts of the ball have the same velocity (magnitude and direction) and the velocity is constant
(4) The centre of the ball moves with constant velocity and the ball spins about its centre uniformly

16. No force acts on particle, it is found that particle has non-zero acceleration. Then :-
(1) Frame of reference is inertial
(2) Frame of reference is non-inertial
(3) Newton's first law is applicable in this situation
(4) None of the above

17. Action and reaction are equal and opposite. If F be the magnitude of both the action and reaction, then :-
(1) Resultant of action and reaction is zero on a body.
(2) Resultant of action and reaction is 2F
(3) Resultant of action and reaction is less than F
(4) Action and reaction do not cancel each-other

18. A box is placed on the surface of a truck. When the truck accelerates in the forward direction, then the direction of force of friction on lower surface of the box due to surface of the truck to prevent the sliding of the box is :-
(1) In backward direction
(2) In forward direction
(3) In upward direction
(4) In downward direction

19. A car accelerates on a horizontal road due to the force exerted by :-
(1) The road
(2) The driver of car
(3) The engine of the car
(4) The tyre

20. A hockey player is moving northward and suddenly turns westward with the same speed to avoid an opponent. The force that acts on the player is :-
(1) Frictional force along westward
(2) Muscle force along southward
(3) Frictional force along south-west
(4) Muscle force along south-west

21. A train is accelerating with constant acceleration 'a' on a straight track. If a ball of mass 'm' is dropped outside from the window of train, then magnitude of net acceleration of ball, with respect to observer situated at ground, will be (neglect air resistance)
(1) a (2) g
(3) g + a (4) $\sqrt{g^2+a^2}$

22. **Statement-1 :** If we consider system of two bodies A and B as a whole, F_{AB} and F_{BA} are internal forces of the system (A+B). They add to give a null force.
Statement-2 : Internal forces in a body or a system of particles cancel away in pairs.
Correct statements is/are :
(1) Only 1 (2) Only 2
(3) Both 1 and 2 (4) Neither 1 nor 2

23. If two bullets of same mass (m) thrown with different speeds v_1 and v_2 ($v_1 > v_2$) then which bullet damage more ?
(1) First bullet
(2) Second bullet
(3) Same damage by both
(4) None of these

24. If no external force acts on particle, then which of the following is incorrect about particle ?
(1) Particle may be at rest
(2) Particle moves with uniform speed on linear path
(3) Particle moves with uniform speed on circle
(4) None of these

25. Which of the following statements is not true ?
(1) The same force is required to act for same time for the same change in momentum for different bodies of same mass
(2) The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction in which the force acts
(3) A greater opposing force is needed to stop a heavy body than a light body in the same time, if they are moving with the same speed
(4) The greater is the change in the momentum in a given time, the lesser is the force that needs to be applied

26. μ_s, μ_k and μ_r are the coefficients of static friction, kinetic friction and rolling friction respectively then :-
(1) $\mu_s > \mu_k > \mu_r$
(2) $\mu_s > \mu_k < \mu_r$
(3) $\mu_s < \mu_k > \mu_r$
(4) $\mu_s < \mu_k < \mu_r$

27. Which type of forces does a proton exert o another proton?
(1) Nuclear
(2) Electromagnetic, nuclear
(3) Gravitation, Electromagnetic, Nuclear
(4) Gravitation, Nuclear

28. Which of the following statement is not true fo a force?
(1) Force can produce deformation in a body
(2) Force is the influence of one body on anothe
(3) Force can produce acceleration in a body
(4) Force always acts in the direction of motio

29. Identify the incorrect statement of the following:

(1) The weight of a body always acts towards the centre of the earth

(2) The contact force between the two surfaces may be at angle with surface

(3) The tension in a string always pulls a body

(4) The spring force always pushes the body

30. A body is said to be in equilibrium:

(1) If it is at rest

(2) If it is moving with constant speed

(3) If it is not accelerating

(4) If even number of forces are acting on the body

31. Inertia of a body is defined as:

(1) Its ability to remain at rest

(2) Its ability to get removed

(3) Its ability to oppose the changes in its translatory state of motion

(4) All of the above

32. According to Newton's third law:

(1) Forces exist in pairs which are equal and opposite

(2) The action and reaction forces are always of the same nature

(3) The action and reaction always act on different bodies

(4) All of these

33. Identify the incorrect statement related to the static and kinetic friction between the two bodies:

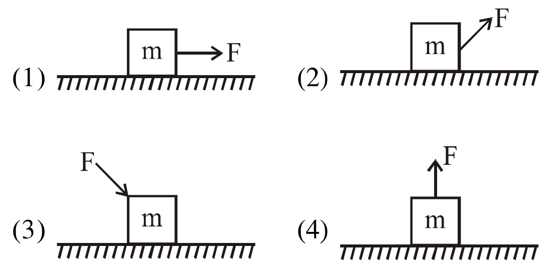
(1) The magnitude of static friction between the two surfaces is always greater than the kinetic friction

(2) The magnitude of kinetic friction is always of a constant magnitude

(3) The magnitude of static friction lies between zero and a maximum value

(4) The coefficient of static friction is always more than the coefficient of kinetic friction

34. A block of mass m is placed at rest on a rough horizontal surface with the coefficient of friction μ. The following figure shows four different ways in which a constant force F may be applied on the block. Identify the situation in which the block can be moved with the least effort:



35. A particle is moving with a constant speed along a straight line path. A force is not required to:

(1) increase its speed

(2) decrease its momentum

(3) change the direction

(4) keep it moving with uniform velocity

36. An object will continue accelerating until

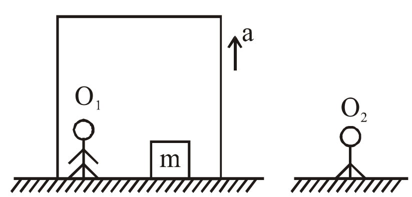
(1) Resultant force on it begins to decrease

(2) Its velocity changes direction

(3) The resultant force on it is zero

(4) The resultant force is at right angles to its direction of motion

37. Observer O_1 is in a lift going upward and O_2 is on ground both apply Newton's law and measure normal reaction on the body:



(1) The both measure the same value

(2) The both measure different value

(3) The both measure zero

(4) Not sufficient data

38. Neglect the effect of the rotation of the Earth. Suppose the Earth suddenly stops attracting objects placed near its surface. A person standing on the surface of the earth will:

(1) Fly up

(2) Slip along the surface

(3) Fly along a tangent to the earth's surface

(4) Remain standing

39. A cold soft drink is kept on the balance. When the cap is open, then the weight:

(1) Increases

(2) Decreases

(3) First increases then decreases

(4) Remains same

40. A student attempts to pull himself up by tugging on his hair. He will not succeed:

(1) As the force exerted is small

(2) The frictional force while gripping is small

(3) Newton's law of Inertia is not applicable to living beings

(4) As the force applied is internal to the system

41. Two ends of a spring are displaced along the length of the spring. All displacements have equal magnitude. In which case(s) the tension or compression in the spring will have a maximum magnitude?

(A) The right end is displaced towards right and the left end towards left.

(B) Both ends are displaced towards right.

(C) Both ends are displaced towards left.

(D) The right end is displaced towards left and the left end towards right.

(1) A and D (2) B and C

(3) All cases (4) None

42. Four forces act on a point object. The object will be in equilibrium, if :-

(A) all of them are in the same plane.

(B) they are opposite to each other in pairs.

(C) the sum of x, y and z components of forces is zero separately.

(D) they form a closed figure of 4 sides when added as per polygon law.

(1) a, b, c (2) a, c, d

(3) b, c, d (4) a, b, d

43. A block is placed on the top of a smooth inclined plane of inclination θ kept on the floor of a lift. When the lift in descending with retardation 'a', the block is released. The acceleration of the block relative to the incline is :-

(1) $g\sin\theta$ (2) $a\sin\theta$

(3) $(g-a)\sin\theta$ (4) $(g+a)\sin\theta$

44. In non-intertial frame, the second law of motion is written as :-

(1) $F = ma$ (2) $F = ma + F_P$

(3) $F = ma - F_P$ (4) $F = 2ma$

(Where F_P is pseudo-force, while a is the acceleration of the body relative to non-inertial frame)

45. If non external force acts on particle, then which of the following is incorrect about particle?

(1) Particle may be at rest

(2) Particle moves with uniform velocity on linear path

(3) Particle moves with uniform speed on circle

(4) None of the above